

### 3. Methodology

Although YOLOv8 achieves an excellent balance between accuracy and speed for generic object detection, it suffers from three major limitations when applied directly to pavement distress detection, especially for elongated cracks and diversified regional defects (e.g., potholes and patching areas). First, the standard convolution and C2f blocks are not sensitive enough to high-frequency, directional features of cracks, and the multi-scale receptive field design is monolithic. Second, the IoU-based regression loss is highly sensitive to small positional offsets and produces zero gradients when predicted and ground-truth boxes do not overlap; moreover, it fails to exploit the morphological priors (e.g., aspect ratio, area) that distinguish line-like cracks from regional distresses. Third, conventional non-maximum suppression (NMS) often retains multiple fragmented boxes belonging to the same continuous crack because their pairwise IoU is low, leading to duplicate detections.

To address these issues, we propose a **box-supervised morphology-aware lightweight YOLO model (BMS-YOLO)** that requires only bounding-box annotations. The methodological innovations include three aspects: a morphology-aware feature extraction backbone that integrates FDDE, MorphSparseMoE and ECA; a box-driven morphological consistency loss consisting of WIoU and a morphology term; and a context-aware confidence calibration followed by topology-guided box fusion at inference.

#### 3.1 Morphology-Aware Feature Extraction Backbone

The original YOLOv8 employs the C2f module as its main building block. For an input feature map  $X \in \mathbb{R}^{C \times H \times W}$ , C2f splits the output of a  $1 \times 1$  convolution into two branches, passes one branch through a chain of  $n$  standard Bottlenecks (each consisting of two  $3 \times 3$  convolutions with a residual connection), and concatenates all intermediate results. Although computationally efficient, this design suffers from three shortcomings in crack detection: (i) the standard  $3 \times 3$  convolution is isotropic and cannot effectively capture elongated, line-like structures; (ii) the fixed receptive field does not adapt to the wide variety of crack scales (from very thin hairline cracks to wide spalls); (iii) regional defects (potholes, patches) lack a dedicated low-frequency pathway that preserves smooth intensity variations.

To overcome these limitations, we redesign the backbone with three core components: **FDDE** (Frequency-Direction Detail Enhancement), **MorphSparseMoE** (Morphology-Aware Sparse Mixture of Experts), and **ECA** (Efficient Channel Attention). These components are embedded into a novel **BMS-C2f** block and a **LightSPPFC-SPC** block, as described below.

### 3.1.1 FDDE: Frequency-Direction Detail Enhancement

Cracks are predominantly high-frequency signals (sharp edges), while regional distresses are low-frequency (smooth intensity variations). We propose the FDDE module to explicitly decompose features into high- and low-frequency components and enhance them separately.

Given an input feature map  $X \in \mathbb{R}^{C \times H \times W}$ , the low-frequency component is first obtained by average pooling with a kernel size of  $3 \times 3$ , stride 1, and padding 1:

$$X_{\text{low}} = \text{AvgPool}_{3 \times 3}(X).$$

The high-frequency component is then computed as the residual:

$$X_{\text{high}} = X - X_{\text{low}}.$$

The high-frequency path uses two directional depthwise convolutions to emphasize crack orientations. Depthwise convolution operates on each input channel independently, which keeps the parameter count low. Specifically:

$$\begin{aligned} X_{\text{high}}^{(h)} &= \text{DWConv}_{1 \times 5}(X_{\text{high}}), \\ X_{\text{high}}^{(v)} &= \text{DWConv}_{5 \times 1}(X_{\text{high}}), \\ X'_{\text{high}} &= \text{Conv}_{1 \times 1}(X_{\text{high}}^{(h)} + X_{\text{high}}^{(v)}). \end{aligned}$$

The low-frequency path is deliberately kept simple, using only a  $1 \times 1$  convolution to adjust channel dimensions:

$$X'_{\text{low}} = \text{Conv}_{1 \times 1}(X_{\text{low}}).$$

To adaptively balance the contribution of high-frequency details and low-frequency context, we compute a spatial gate signal from the high-frequency features. The gate is generated by global average pooling followed by a  $1 \times 1$  convolution and a sigmoid activation:

$$g = \sigma\left(\text{Conv}_{1 \times 1}(\text{GAP}(X'_{\text{high}}))\right) \in \mathbb{R}^{C \times 1 \times 1},$$

where  $\sigma(\cdot)$  is the sigmoid function and  $\text{GAP}(\cdot)$  denotes global average pooling that reduces the spatial dimensions to  $1 \times 1$ . The gate value lies in  $(0, 1)$  and is broadcast across the spatial dimensions. The two branches are then combined via a weighted sum and a residual connection:

$$X_{\text{fused}} = X + \text{Conv}_{1 \times 1}\left([g \cdot X'_{\text{high}}, (1 - g) \cdot X'_{\text{low}}]\right),$$

where  $[\cdot, \cdot]$  denotes channel concatenation. The final  $1 \times 1$  convolution reduces the channel dimension from  $2C$  back to  $C$ . This design adds negligible computational cost (approximately 2% more FLOPs) while significantly improving the model's ability to preserve fine crack edges and suppress irrelevant low-frequency noise.

**Advantages of FDDE:** (i) It explicitly separates high- and low-frequency information, avoiding the implicit mixing in standard convolutions. (ii) The directional depthwise convolutions are lightweight yet highly effective for cracks. (iii) The adaptive gate allows the model to decide, per spatial location and per channel, whether to emphasize edges or smooth regions.

### 3.1.2 MorphSparseMoE: Morphology-Aware Sparse Mixture of Experts

To handle the diversity of defect shapes (line-like cracks, isotropic spalls, regional potholes), we introduce a sparse mixture-of-experts (MoE) block with four specialized experts. Each expert is a lightweight sequential module consisting of a spatial operation followed by a  $1 \times 1$  projection. Table 1 summarizes the experts.

表 1: Specialized experts in MorphSparseMoE

| Expert | Operation                                      | Purpose                                        |
|--------|------------------------------------------------|------------------------------------------------|
| $E_0$  | DWConv $_{1 \times 5}$ + Conv $_{1 \times 1}$  | Horizontal line-like cracks                    |
| $E_1$  | DWConv $_{5 \times 1}$ + Conv $_{1 \times 1}$  | Vertical line-like cracks                      |
| $E_2$  | DWConv $_{3 \times 3}$ + Conv $_{1 \times 1}$  | Isotropic small defects (spalls, short cracks) |
| $E_3$  | AvgPool $_{3 \times 3}$ + Conv $_{1 \times 1}$ | Regional context (potholes, patches)           |

Given an input  $X \in \mathbb{R}^{C \times H \times W}$ , a routing network computes expert logits. The routing network first applies global average pooling to obtain a channel-wise descriptor, then uses two fully-connected layers with a SiLU activation in between:

$$r = W_2 \cdot \text{SiLU}(W_1 \cdot \text{GAP}(X)) \in \mathbb{R}^{N_{\text{exp}}},$$

where  $W_1 \in \mathbb{R}^{(C/4) \times C}$ ,  $W_2 \in \mathbb{R}^{N_{\text{exp}} \times (C/4)}$ , and  $N_{\text{exp}} = 4$ . The routing weights are obtained by softmax normalization:

$$w_i = \frac{\exp(r_i)}{\sum_{j=1}^{N_{\text{exp}}} \exp(r_j)}.$$

To encourage sparsity and reduce computational cost, we keep only the top- $k$  experts (in our experiments  $k = 2$ ). The sparse weights are:

$$w_i^{\text{sparse}} = \begin{cases} w_i / \sum_{j \in \text{topk}} w_j, & i \in \text{topk}, \\ 0, & \text{otherwise.} \end{cases}$$

The expert outputs are combined linearly:

$$X_{\text{moe}} = \sum_{i=0}^{N_{\text{exp}}-1} w_i^{\text{sparse}} \cdot E_i(X).$$

Finally, a residual connection and a  $1 \times 1$  projection are applied:

$$X_{\text{out}} = X + \text{Conv}_{1 \times 1}(X_{\text{moe}}).$$

**Advantages of MorphSparseMoE:** (i) It provides morphological specialization without exploding the parameter count because experts are lightweight and only top- $k$  are activated. (ii) The routing network is data-driven and adapts to the local image content, allowing the model to use horizontal/vertical experts for cracks and the isotropic or regional experts for other distress types. (iii) The sparse activation pattern naturally regularizes the model and improves inference speed.

### 3.1.3 Integration into BMS-C2f and LightSPPFCSPC

We replace the original Bottleneck inside each C2f block with a new **BMSBottleneck**:

$$\text{BMSBottleneck}(X) = \text{Conv}_{3 \times 3} \left( \text{MorphSparseMoE}(\text{FDDE}(\text{Conv}_{1 \times 1}(X))) \right) + X,$$

where the shortcut connection is applied when the input and output channel numbers match; otherwise, a  $1 \times 1$  projection is used for the residual path.

The **BMSC2f** block follows the same topology as the original C2f but uses BMSBottleneck. After the final concatenation, we attach an **ECA** (Efficient Channel Attention) module:

$$\text{ECA}(X) = X \cdot \sigma \left( \text{Conv1D}_k(\text{GAP}(X)) \right),$$

where the 1D convolution kernel size is adaptively set as  $k = \frac{\lfloor \log_2(C) + 1 \rfloor}{2}$  (forced to odd). ECA captures cross-channel interactions with virtually zero parameter overhead.

At the top of the neck, we replace the standard SPPF with a **LightSPPFCSPC** block. Its design is:

$$\begin{aligned} X_1 &= \text{Conv}_{1 \times 1}(X), \quad X_2 = \text{Conv}_{1 \times 1}(X), \\ Y_1 &= \text{Conv}_{3 \times 3}(X_1), \\ Y_1^{(0)} &= Y_1, \quad Y_1^{(1)} = \text{MaxPool}_{5 \times 5}(Y_1), \quad Y_1^{(2)} = \text{MaxPool}_{5 \times 5}(Y_1^{(1)}), \quad Y_1^{(3)} = \text{MaxPool}_{5 \times 5}(Y_1^{(2)}), \\ Y_1^{\text{cat}} &= \text{Conv}_{1 \times 1}([Y_1^{(0)}, Y_1^{(1)}, Y_1^{(2)}, Y_1^{(3)}]), \\ X_{\text{out}} &= \text{ECA} \left( \text{Conv}_{1 \times 1}([Y_1^{\text{cat}}, X_2]) \right). \end{aligned}$$

This design progressively enlarges the receptive field while keeping the parameter count low.

## 3.2 Box-Driven Morphological Consistency Loss

The default CIoU loss in YOLOv8 is suboptimal for crack detection for three reasons: (i) small positional offsets cause drastic IoU drops (e.g., a  $6 \times 6$  pixel crack shifted by 1 pixel reduces IoU from 0.53 to 0.06), leading to unstable positive/negative sample assignment;

(ii) when the predicted box and the ground-truth box have no overlap, the IoU is zero and the gradient vanishes, making it impossible to optimize those samples; (iii) CIoU does not explicitly enforce geometric priors such as aspect ratio and area, which are crucial for distinguishing line-like cracks from regional distresses.

To address these issues, we design a novel box loss that combines a distance-aware IoU variant (WIoU) and a morphology consistency term that only requires bounding-box annotations.

### 3.2.1 WIoU: Distance-Aware Dynamic Focusing Loss

We adopt a Wise-IoU (WIoU) style loss that introduces a dynamic focusing factor based on the center distance. Let  $B_p = (x_p, y_p, w_p, h_p)$  be the predicted box (center coordinates and width/height) and  $B_t = (x_t, y_t, w_t, h_t)$  be the ground-truth box. The standard IoU is defined as:

$$\text{IoU} = \frac{|B_p \cap B_t|}{|B_p \cup B_t|}.$$

The Euclidean distance between the box centers is:

$$d = \sqrt{(x_p - x_t)^2 + (y_p - y_t)^2}.$$

Let  $D$  be the diagonal length of the smallest axis-aligned rectangle that encloses both boxes:

$$D = \sqrt{(\max(x_p + w_p/2, x_t + w_t/2) - \min(x_p - w_p/2, x_t - w_t/2))^2 + (\max(y_p + h_p/2, y_t + h_t/2) - \min(y_p - h_p/2, y_t - h_t/2))^2}.$$

Our WIoU loss is then:

$$\mathcal{L}_{\text{WIoU}} = (1 - \text{IoU}) \cdot \exp\left(\frac{d^2}{D^2}\right).$$

Importantly, the exponential term  $\exp(d^2/D^2)$  is detached from the gradient computation (i.e., treated as a constant weight). This ensures that the loss gradient is proportional to the gradient of  $(1 - \text{IoU})$  scaled by a positive factor, avoiding any distortion of the gradient direction. The weight  $\exp(d^2/D^2)$  is larger when the center deviation is large, thus forcing the model to focus on samples with poor localization.

**Advantages of WIoU:** (i) It reduces the influence of small positional jitter on the loss value because the weight penalizes large deviations more heavily. (ii) It maintains non-zero gradients even when IoU is zero, as long as the boxes are not exactly overlapping (the weight still exists). (iii) The dynamic weighting improves the model's convergence speed by emphasizing hard examples.

### 3.2.2 Morphology Consistency Loss

To further exploit the morphological priors that are specific to cracks, we add two auxiliary loss terms that operate only on bounding-box coordinates. These terms do not require pixel-wise segmentation masks.

Let  $w_p, h_p$  be the width and height of the predicted box, and  $w_t, h_t$  those of the ground-truth box. The **aspect-ratio consistency loss** is defined in the log space to treat over-estimation and under-estimation symmetrically:

$$\mathcal{L}_{\text{aspect}} = \text{SmoothL1}\left(\left|\log \frac{w_p}{h_p} - \log \frac{w_t}{h_t}\right|\right).$$

The log ratio makes the loss invariant to the absolute scale of the box. For a line-like crack,  $\log(w/h)$  is large in magnitude (positive for horizontal cracks, negative for vertical cracks, but we take absolute value to ignore orientation). The SmoothL1 function is defined as:

$$\text{SmoothL1}(x) = \begin{cases} 0.5x^2, & |x| < 1, \\ |x| - 0.5, & \text{otherwise.} \end{cases}$$

The **area consistency loss** is defined similarly:

$$\mathcal{L}_{\text{area}} = \text{SmoothL1}\left(\left|\log \sqrt{w_p h_p} - \log \sqrt{w_t h_t}\right|\right).$$

Here  $\sqrt{wh}$  represents the geometric mean of the side lengths, which is proportional to the square root of the area. Using the log ensures that small and large boxes contribute equally.

The overall morphology loss is a weighted sum:

$$\mathcal{L}_{\text{morph}} = \mathcal{L}_{\text{aspect}} + 0.5 \mathcal{L}_{\text{area}}.$$

The coefficient 0.5 is chosen empirically to give slightly more importance to aspect ratio, which is more discriminative for cracks.

Crucially, this morphology loss is applied **only to crack classes**. We automatically identify crack categories from the dataset label names by matching keywords such as “crack”, “D00”, “D10”, “D20”, “裂”, etc. For non-crack classes (e.g., potholes, repairs), the morphology loss is disabled because their aspect ratio can vary widely and imposing an elongation prior would be harmful. The total box loss is:

$$\mathcal{L}_{\text{box}} = \mathcal{L}_{\text{WIoU}} + \lambda_{\text{morph}} \cdot \mathcal{L}_{\text{morph}},$$

with  $\lambda_{\text{morph}} = 0.02$  in all experiments. This small weight ensures that the morphology loss acts as a mild regularizer rather than dominating the optimization.

**Advantages of the morphology consistency loss:** (i) It injects domain knowledge about crack geometry using only weak box supervision, which is much cheaper to annotate than segmentation masks. (ii) The log-space formulation makes the loss scale-invariant, which is important because cracks can appear at vastly different scales (from a few pixels to hundreds of pixels). (iii) By applying it only to crack classes, we avoid interfering with the learning of other distress types.

### 3.3 Context-Aware Post-processing and Topology-Guided Fusion

Standard NMS suppresses redundant boxes based solely on classification confidence and an IoU threshold (typically 0.5–0.7). For cracks, this approach fails when a single physical crack is fragmented into several small detections with pairwise IoU below the threshold. Moreover, the raw classification confidence often does not reflect the actual presence of crack-like edge structures inside the box. We propose two complementary post-processing steps: confidence calibration using local image statistics, followed by topology-guided box fusion.

#### 3.3.1 Confidence Calibration Using Local Edge Density

For a detection box  $B$  with coordinates  $(x_1, y_1, x_2, y_2)$ , we crop the corresponding region from the original image and convert it to grayscale. Let  $I$  be the grayscale patch. We compute the gradient magnitude map  $G$  using the Sobel operator:

$$G_x = \text{Sobel}_x(I), \quad G_y = \text{Sobel}_y(I), \quad G = |G_x| + |G_y|.$$

We then compute two statistics over all pixels inside the box: - **Density**: the proportion of pixels whose gradient magnitude exceeds the 75-th percentile  $t_{75}$  of  $G$  within the box:

$$\text{density} = \frac{|\{p : G(p) > t_{75}\}|}{\text{area}(B)}.$$

- **Contrast**: the mean gradient magnitude, normalized by an empirical maximum (set to 48 based on statistics of our dataset):

$$\text{contrast} = \min \left( 1, \frac{\text{mean}(G)}{48} \right).$$

The edge credibility score is a weighted combination:

$$\text{edge} = \min(1, 0.6 \cdot (\text{density} \cdot 4) + 0.4 \cdot \text{contrast}).$$

The factor 4 in the density term compensates for the fact that density is typically around 0.1–0.3; after multiplication, it becomes comparable to contrast.

Next, we compute a shape-based score. For a box with width  $w = x_2 - x_1$  and height  $h = y_2 - y_1$ , the aspect ratio is  $\alpha = \max(w/h, h/w)$ . For **crack classes** (elongated), we define:

$$\text{shape} = \text{clamp}\left(\frac{\log(1 + \alpha)}{\log(12)}, 0.35, 1.25\right).$$

The base of the logarithm is chosen so that  $\alpha = 12$  gives  $\text{shape} \approx 1.0$ . The clamping prevents extreme values. The calibration factor is then:

$$f_{\text{crack}} = 0.75 + 0.35 \text{ edge} + 0.15 \text{ shape}.$$

For **non-crack classes** (compact, such as potholes), we use a compactness score:

$$\text{compact} = \text{clamp}\left(\frac{1.5}{\alpha}, 0.55, 1.0\right),$$

and the calibration factor is:

$$f_{\text{non}} = 0.85 + 0.25 \text{ compact} + 0.15 \text{ edge}.$$

The final calibrated confidence is:

$$\text{conf}_{\text{new}} = \text{clamp}(\text{conf}_{\text{orig}} \cdot f, 0.001, 0.999).$$

Boxes whose calibrated confidence falls below a user-defined threshold (e.g., 0.25) are discarded.

**Advantages of confidence calibration:** (i) It leverages low-level image features (gradients) to verify whether the box actually contains a crack-like edge pattern, suppressing false positives from plain backgrounds or smooth patches. (ii) The shape term integrates the expected elongation of cracks, further boosting true positives and reducing spurious detections. (iii) The entire calibration is purely heuristic, requires no extra training, and adds negligible runtime.

### 3.3.2 Topology-Guided Box Fusion (TGF)

Even after NMS and confidence calibration, a long continuous crack may still be represented by several disjoint boxes because the detector’s confidence dips in low-contrast regions. These boxes typically have low IoU (often  $< 0.2$ ) and would not be merged by standard NMS. To address this, we propose a graph-based fusion algorithm that uses a relaxed connectivity criterion specifically designed for cracks.

Let  $\mathcal{B} = \{B_i\}_{i=1}^M$  be the set of boxes that survive the confidence calibration. We construct a Union-Find (disjoint-set) data structure over  $\mathcal{B}$ . Two boxes  $B_i$  and  $B_j$  are considered **connected** if any of the following criteria holds:



- **Criterion A (crack-specific low-IoU):** both boxes belong to a crack class and  $\text{IoU}(B_i, B_j) > 0.12$ . The threshold 0.12 is much lower than the standard NMS threshold, allowing merging of slightly overlapping fragments.
- **Criterion B (orientation-based proximity):** both boxes are crack class, have the same orientation (horizontal if  $w \geq h$ , vertical otherwise), and the Euclidean distance between their centers  $d_{ij}$  satisfies:

$$d_{ij} < \max(0.08 \cdot D_{\text{img}}, 0.65 \cdot \max(L_i, L_j)),$$

where  $D_{\text{img}}$  is the image diagonal length and  $L_i$  is the longer side of box  $B_i$ . This criterion captures fragments that are aligned along the same direction but do not overlap.

- **Criterion C (non-crack default):** both boxes belong to a non-crack class and  $\text{IoU} > 0.35$  (standard threshold).

Once the connectivity graph is built, we identify connected components. For each component  $C = \{B_{i_1}, B_{i_2}, \dots, B_{i_m}\}$ , we fuse the boxes into a single detection as follows:

- **Fused coordinates:** we compute a weighted average of the box coordinates (with weights equal to the calibrated confidences) and also compute the axis-aligned bounding envelope (min/max of all corners). The final coordinates are a convex combination:

$$\begin{aligned} \text{xy}_w &= \frac{\sum_{k=1}^m \text{conf}_k \cdot (x_{1,k}, y_{1,k}, x_{2,k}, y_{2,k})}{\sum_{k=1}^m \text{conf}_k}, \\ \text{xy}_e &= \left( \min_k x_{1,k}, \min_k y_{1,k}, \max_k x_{2,k}, \max_k y_{2,k} \right), \\ \text{xy}_{\text{fused}} &= 0.35 \cdot \text{xy}_w + 0.65 \cdot \text{xy}_e. \end{aligned}$$

The envelope term (65% weight) ensures that the fused box covers the entire span of the crack, while the weighted term (35%) provides a more accurate center when fragments have high confidence.

- **Fused confidence:** we take the mean confidence of the component and add a small reward for multiple fragments:

$$\text{conf}_{\text{fused}} = \text{clamp} \left( \frac{1}{m} \sum_{k=1}^m \text{conf}_k + 0.05 \cdot \ln(1 + m), 0.001, 0.999 \right).$$

The logarithmic reward encourages merging because a crack that is detected in several fragments is likely to be a true positive.

- **Fused class:** the class label of any box in the component (by construction they all belong to the same class; if a conflict arises, we take the class with the highest total confidence).

After fusing all components, the output is the set of fused boxes together with any unmerged single boxes. This fused set replaces the original detections.

**Advantages of topology-guided box fusion:** (i) It explicitly models the fact that a single physical crack can be fragmented in the detection output, which is a common failure mode of CNN-based detectors for thin, elongated objects. (ii) The relaxed connectivity criteria (low IoU and orientation-based distance) are tailored to cracks, while non-crack objects fall back to standard NMS. (iii) The fusion strategy preserves the overall extent of the crack and produces a single, coherent detection, which is much more useful for downstream applications (e.g., crack length estimation).

### 3.4 Training and Inference Settings

All models are trained using SGD with momentum 0.937, weight decay 0.0005, initial learning rate 0.003, cosine annealing schedule, batch size 64 (or 8 on limited GPUs), and 300 epochs. Data augmentation includes mosaic (probability 0.6), mixup (0.05), random scaling ( $\pm 40\%$ ), and HSV shifts. Early stopping with patience 80 is applied. The input image size is  $640 \times 640$ . All experiments are conducted on a single NVIDIA Quadro P2000 (or similar) and PyTorch 1.13. Detailed hyperparameters are listed in the supplementary material.

### 3. 方法

尽管 YOLOv8 在通用目标检测上取得了良好的精度与速度平衡，但将其直接应用于路面病害检测（特别是细长裂纹以及坑洞、修补区等多样化区域病害）时，存在三方面主要局限：标准卷积与 C2f 模块对裂纹的高频方向性特征提取能力不足，且多尺度感受野设计单一；基于 IoU 的回归损失对小尺寸或微小位置偏移的裂纹框极其敏感，且在预测框与真值框无重叠时梯度为零，同时未能利用长宽比、面积等区分线型裂纹与区域病害的形态先验；常规非极大值抑制（NMS）仅依赖类别置信度和 IoU 阈值，对于一条连续但因遮挡或低对比度而产生多个低 IoU 碎片框的裂纹，无法将其合并，导致重复检测。

为克服上述问题，本文提出一种仅依赖边界框标注的形态感知轻量型 YOLO 模型（BMS-YOLO）。方法创新体现在三个方面：构建融合 FDDE、MorphSparseMoE 与 ECA 的形态感知特征提取骨架；设计由 WIoU 与形态一致性损失组成的边界框驱动损失函数；在推理阶段引入上下文感知的置信度校准与拓扑引导的框融合。

#### 3.1 形态感知的特征提取骨架

原始 YOLOv8 使用 C2f 模块作为其基本构件。对于输入特征图  $X \in \mathbb{R}^{C \times H \times W}$ ，C2f 将一个  $1 \times 1$  卷积的输出分割为两支，其中一支经过  $n$  个标准 Bottleneck（每个包含两个  $3 \times 3$  卷积及残差连接）的链式处理，最后将所有中间结果拼接。尽管该设计计算高效，但在裂纹检测中存在三个缺陷：(i) 标准  $3 \times 3$  卷积是各向同性的，无法有效捕获细长、线状结构；(ii) 固定的感受野无法适应从极细裂缝到较宽剥落区的多种裂纹尺度；(iii) 区域型病害（坑洞、修补区）缺少专用的低频路径以保留平缓的强度变化。

为了克服这些局限，我们重新设计骨架网络，核心包含三个子模块：FDDE（频率 - 方向细节增强）、MorphSparseMoE（形态感知稀疏专家混合）和 ECA（高效通道注意力），并将它们嵌入到新的 BMS-C2f 模块与 LightSPPFCSPC 模块中，具体描述如下。

##### 3.1.1 FDDE: 频率 - 方向细节增强模块

裂纹主要表现为高频信号（锐利边缘），而区域病害则为低频信号（平缓强度变化）。我们提出 FDDE 模块，显式地将特征分解为高频和低频分量并分别增强。

对于输入特征图  $X \in \mathbb{R}^{C \times H \times W}$ ，首先通过核大小为  $3 \times 3$ 、步长 1、填充 1 的平均池化得到低频分量：

$$X_{\text{low}} = \text{AvgPool}_{3 \times 3}(X).$$

高频分量则计算为残差：

$$X_{\text{high}} = X - X_{\text{low}}.$$

高频路径使用两个方向性的深度卷积以强调裂纹走向。深度卷积在每个输入通道上独立操作，从而保持较低的参数量。具体地：

$$\begin{aligned} X_{\text{high}}^{(h)} &= \text{DWConv}_{1 \times 5}(X_{\text{high}}), \\ X_{\text{high}}^{(v)} &= \text{DWConv}_{5 \times 1}(X_{\text{high}}), \\ X'_{\text{high}} &= \text{Conv}_{1 \times 1}(X_{\text{high}}^{(h)} + X_{\text{high}}^{(v)}). \end{aligned}$$

低频路径刻意保持简单，仅使用一个  $1 \times 1$  卷积来调整通道维度：

$$X'_{\text{low}} = \text{Conv}_{1 \times 1}(X_{\text{low}}).$$

为了自适应地平衡高频细节和低频上下文的贡献，我们基于高频特征计算一个空间门控信号。门控通过全局平均池化后接  $1 \times 1$  卷积和 Sigmoid 激活生成：

$$g = \sigma\left(\text{Conv}_{1 \times 1}(\text{GAP}(X'_{\text{high}}))\right) \in \mathbb{R}^{C \times 1 \times 1},$$

其中  $\sigma(\cdot)$  是 Sigmoid 函数， $\text{GAP}(\cdot)$  表示全局平均池化，将空间维度降为  $1 \times 1$ 。门控值位于  $(0, 1)$  之间，并在空间维度上广播。两个分支通过加权和与残差连接组合：

$$X_{\text{fused}} = X + \text{Conv}_{1 \times 1}\left([g \cdot X'_{\text{high}}, (1 - g) \cdot X'_{\text{low}}]\right),$$

其中  $[\cdot, \cdot]$  表示通道拼接。最后的  $1 \times 1$  卷积将通道数从  $2C$  降回  $C$ 。该设计计算开销极小（大约增加 2% FLOPs），同时显著提升了模型保留细裂纹边缘和抑制无关低频噪声的能力。

**FDDE 的优势：** (i) 显式分离高频与低频信息，避免了标准卷积中的隐式混合。(ii) 方向性深度卷积轻量且对裂纹非常有效。(iii) 自适应门控允许模型在每个空间位置和每个通道上决策是增强边缘还是平滑区域。

### 3.1.2 MorphSparseMoE：形态感知稀疏专家混合

为了处理多样的病害形状（线状裂纹、各向同性剥落、区域型坑洞），我们引入一个稀疏混合专家（MoE）块，包含四个专用专家。每个专家为一个轻量序列：一个空间操作后接一个  $1 \times 1$  投影。表 1 总结了各专家。

表 2: MorphSparseMoE 中的专用专家

| 专家    | 操作                                                       | 用途              |
|-------|----------------------------------------------------------|-----------------|
| $E_0$ | $\text{DWConv}_{1 \times 5} + \text{Conv}_{1 \times 1}$  | 水平线状裂纹          |
| $E_1$ | $\text{DWConv}_{5 \times 1} + \text{Conv}_{1 \times 1}$  | 垂直线状裂纹          |
| $E_2$ | $\text{DWConv}_{3 \times 3} + \text{Conv}_{1 \times 1}$  | 各向同性小缺陷（剥落、短裂纹） |
| $E_3$ | $\text{AvgPool}_{3 \times 3} + \text{Conv}_{1 \times 1}$ | 区域上下文（坑洞、修补区）   |

对于输入  $X \in \mathbb{R}^{C \times H \times W}$ ，路由网络计算专家 logits。路由网络首先应用全局平均池化获得通道描述符，然后使用两个全连接层，中间带有 SiLU 激活：

$$r = W_2 \cdot \text{SiLU}(W_1 \cdot \text{GAP}(X)) \in \mathbb{R}^{N_{\text{exp}}},$$

其中  $W_1 \in \mathbb{R}^{(C/4) \times C}$ ， $W_2 \in \mathbb{R}^{N_{\text{exp}} \times (C/4)}$ ， $N_{\text{exp}} = 4$ 。路由权重通过 softmax 归一化得到：

$$w_i = \frac{\exp(r_i)}{\sum_{j=1}^{N_{\text{exp}}} \exp(r_j)}.$$

为鼓励稀疏性并减少计算量，我们只保留 top- $k$  专家（实验中  $k = 2$ ）。稀疏权重为：

$$w_i^{\text{sparse}} = \begin{cases} w_i / \sum_{j \in \text{topk}} w_j, & i \in \text{topk}, \\ 0, & \text{其他情况}. \end{cases}$$

专家输出线性组合为：

$$X_{\text{moe}} = \sum_{i=0}^{N_{\text{exp}}-1} w_i^{\text{sparse}} \cdot E_i(X).$$

最后应用残差连接和  $1 \times 1$  投影：

$$X_{\text{out}} = X + \text{Conv}_{1 \times 1}(X_{\text{moe}}).$$

**MorphSparseMoE 的优势：** (i) 提供形态专用化而不导致参数爆炸，因为专家轻量且只有 top- $k$  被激活。(ii) 路由网络是数据驱动的，适应局部图像内容，使模型能够对裂纹使用水平/垂直专家，对其他病害使用各向同性或区域专家。(iii) 稀疏激活模式自然正则化模型并提高推理速度。

### 3.1.3 整合到 BMS-C2f 与 LightSPPFCSPC

我们将每个 C2f 块内的原始 Bottleneck 替换为新的 **BMSBottleneck**。BMSBottleneck 定义为：

$$\text{BMSBottleneck}(X) = \text{Conv}_{3 \times 3}(\text{MorphSparseMoE}(\text{FDDE}(\text{Conv}_{1 \times 1}(X)))) + X,$$

当输入输出通道数匹配时使用残差连接，否则在残差路径中使用  $1 \times 1$  投影。

**BMSC2f** 块保持与原 C2f 相同的拓扑，但使用 BMSBottleneck。在最终的拼接之后，我们附加一个 **ECA**（高效通道注意力）模块：

$$\text{ECA}(X) = X \cdot \sigma(\text{Conv1D}_k(\text{GAP}(X))),$$

其中 1D 卷积核大小自适应设为  $k = \frac{\lfloor \log_2(C) + 1 \rfloor}{2}$ （强制为奇数）。ECA 以几乎为零的参数开销捕获跨通道交互。

在颈部顶层，我们使用 **LightSPPFCSPC** 块替代标准 SPPF。该块的设计为：

$$\begin{aligned}
X_1 &= \text{Conv}_{1 \times 1}(X), \quad X_2 = \text{Conv}_{1 \times 1}(X), \\
Y_1 &= \text{Conv}_{3 \times 3}(X_1), \\
Y_1^{(0)} &= Y_1, \quad Y_1^{(1)} = \text{MaxPool}_{5 \times 5}(Y_1), \quad Y_1^{(2)} = \text{MaxPool}_{5 \times 5}(Y_1^{(1)}), \quad Y_1^{(3)} = \text{MaxPool}_{5 \times 5}(Y_1^{(2)}), \\
Y_1^{\text{cat}} &= \text{Conv}_{1 \times 1}([Y_1^{(0)}, Y_1^{(1)}, Y_1^{(2)}, Y_1^{(3)}]), \\
X_{\text{out}} &= \text{ECA}(\text{Conv}_{1 \times 1}([Y_1^{\text{cat}}, X_2])).
\end{aligned}$$

该设计逐步扩大感受野，同时保持较低的参数量。

## 3.2 边界框驱动的形态一致性损失函数

YOLOv8 默认的 CIoU 损失在裂纹检测中并非最优，原因有三：(i) 小的位置偏移会导致 IoU 急剧下降（例如  $6 \times 6$  像素的裂纹偏移 1 像素，IoU 从 0.53 降至 0.06），导致正负样本分配不稳定；(ii) 当预测框与真值框无重叠时 IoU 为零且梯度消失，这些样本无法优化；(iii) CIoU 没有显式约束长宽比、面积等几何先验，而这些正是区分线型裂纹与区域病害的关键。

为解决这些问题，我们设计了一种新颖的框损失，结合了距离感知的 IoU 变体 (WIoU) 和一个仅需边界框标注的形态一致性项。

### 3.2.1 WIoU：基于中心距离的动态聚焦损失

我们采用 Wise-IoU (WIoU) 风格的损失，引入基于中心距离的动态聚焦因子。设预测框为  $B_p = (x_p, y_p, w_p, h_p)$ （中心坐标及宽高），真值框为  $B_t = (x_t, y_t, w_t, h_t)$ 。标准 IoU 定义为：

$$\text{IoU} = \frac{|B_p \cap B_t|}{|B_p \cup B_t|}.$$

两框中心的欧氏距离为：

$$d = \sqrt{(x_p - x_t)^2 + (y_p - y_t)^2}.$$

令  $D$  为覆盖两个框的最小轴对齐矩形的对角线长度：

$$D = \sqrt{(\max(x_p + w_p/2, x_t + w_t/2) - \min(x_p - w_p/2, x_t - w_t/2))^2 + (\max(y_p + h_p/2, y_t + h_t/2) - \min(y_p - h_p/2, y_t - h_t/2))^2}.$$

我们的 WIoU 损失为：

$$\mathcal{L}_{\text{WIoU}} = (1 - \text{IoU}) \cdot \exp\left(\frac{d^2}{D^2}\right).$$

重要的是，指数项  $\exp(d^2/D^2)$  从梯度计算中阻断（即视为常数权重）。这确保了损失梯度正比于  $(1 - \text{IoU})$  的梯度乘以一个正因子，避免了梯度方向的任何扭曲。权重  $\exp(d^2/D^2)$  在中心偏差大时较大，从而迫使模型关注定位差的样本。

**WIoU 的优势：** (i) 由于权重对大的偏差惩罚更重，减小了小位置抖动对损失值的影响。(ii) 即使 IoU 为零，只要框不完全重合，权重仍然存在，从而保持非零梯度。(iii) 动态加权通过强调困难样本提高了模型收敛速度。

### 3.2.2 形态一致性损失

为了进一步利用裂纹特有的形态先验，我们增加了两个辅助损失项，它们仅作用于边界框坐标，不需要像素级分割掩膜。

设预测框的宽高为  $w_p, h_p$ ，真值框的宽高为  $w_t, h_t$ 。长宽比一致性损失在对数空间中定义，以对称地处理高估和低估：

$$\mathcal{L}_{\text{aspect}} = \text{SmoothL1} \left( \left| \log \frac{w_p}{h_p} - \log \frac{w_t}{h_t} \right| \right).$$

对数比率使损失对框的绝对尺度不变。对于线状裂纹， $\log(w/h)$  的绝对值很大（水平裂纹为正，垂直为负，但取绝对值忽略方向）。SmoothL1 函数定义为：

$$\text{SmoothL1}(x) = \begin{cases} 0.5x^2, & |x| < 1, \\ |x| - 0.5, & \text{其他情况}. \end{cases}$$

面积一致性损失类似地定义：

$$\mathcal{L}_{\text{area}} = \text{SmoothL1} \left( \left| \log \sqrt{w_p h_p} - \log \sqrt{w_t h_t} \right| \right).$$

这里  $\sqrt{wh}$  代表边长几何平均，与面积的平方根成正比。使用对数确保大小框贡献相等。

总的形态损失为加权和：

$$\mathcal{L}_{\text{morph}} = \mathcal{L}_{\text{aspect}} + 0.5 \mathcal{L}_{\text{area}}.$$

系数 0.5 是经验选择，给予长宽比略高的权重，因为其对裂纹更具判别力。

关键地，此形态损失仅应用于裂纹类别。我们通过匹配数据集标签名称中的关键词（如“crack”、“D00”、“D10”、“D20”、“裂”等）自动识别裂纹类别。对于非裂纹类（如坑洞、修补），禁用形态损失，因为它们的长宽比变化很大，施加细长先验会有害。总框损失为：

$$\mathcal{L}_{\text{box}} = \mathcal{L}_{\text{WIoU}} + \lambda_{\text{morph}} \cdot \mathcal{L}_{\text{morph}},$$

所有实验中取  $\lambda_{\text{morph}} = 0.02$ 。这个小的权重确保形态损失仅作为一个温和的正则器，而不是主导优化。

**形态一致性损失的优势：** (i) 仅使用弱框监督注入裂纹几何的领域知识，标注成本远低于分割掩膜。(ii) 对数空间公式使损失尺度不变，这对于裂纹可能出现在从几个像素到数百像素的极不同尺度上很重要。(iii) 仅应用于裂纹类别，避免干扰其他病害类型的学习。

### 3.3 上下文感知的后处理与拓扑融合

标准 NMS 仅基于分类置信度和一个 IoU 阈值（通常 0.5-0.7）抑制冗余框。对于裂纹，当一条物理裂纹被分割成几个小检测片段且它们之间的 IoU 低于阈值时，这种方法会失效。此外，原始分类置信度往往不能反映框内实际存在裂纹状边缘结构。我们提出两个互补的后处理步骤：使用局部图像统计进行置信度校准，然后进行拓扑引导的框融合。

#### 3.3.1 基于局部边缘密度的置信度校准

对于检测框  $B$ （坐标  $(x_1, y_1, x_2, y_2)$ ），我们从原始图像裁剪对应区域并转换为灰度图。令  $I$  为灰度图块。使用 Sobel 算子计算梯度幅度图  $G$ ：

$$G_x = \text{Sobel}_x(I), \quad G_y = \text{Sobel}_y(I), \quad G = |G_x| + |G_y|.$$

然后我们计算框内所有像素的两个统计量：- **密度**：梯度幅度超过框内  $G$  的 75% 分位数  $t_{75}$  的像素比例：

$$\text{density} = \frac{|\{p : G(p) > t_{75}\}|}{\text{area}(B)}.$$

- **对比度**：平均梯度幅度，用经验最大值（根据我们数据集的统计设为 48）归一化：

$$\text{contrast} = \min\left(1, \frac{\text{mean}(G)}{48}\right).$$

边缘可信度分数为加权组合：

$$\text{edge} = \min(1, 0.6 \cdot (\text{density} \cdot 4) + 0.4 \cdot \text{contrast}).$$

密度项中的因子 4 用于补偿密度通常在 0.1-0.3 之间的实际情况；乘以 4 后与对比度可比。

接下来，计算基于形状的分值。对于宽度  $w = x_2 - x_1$  和高度  $h = y_2 - y_1$  的框，长宽比  $\alpha = \max(w/h, h/w)$ 。对于**裂纹类**（细长），我们定义：

$$\text{shape} = \text{clamp}\left(\frac{\log(1 + \alpha)}{\log(12)}, 0.35, 1.25\right).$$

选择对数的底使得  $\alpha = 12$  时  $\text{shape} \approx 1.0$ 。裁剪防止极端值。校准因子为：

$$f_{\text{crack}} = 0.75 + 0.35 \text{ edge} + 0.15 \text{ shape}.$$

对于**非裂纹类**（紧凑，如坑洞），使用紧凑度分数：

$$\text{compact} = \text{clamp}\left(\frac{1.5}{\alpha}, 0.55, 1.0\right),$$

校准因子为：

$$f_{\text{non}} = 0.85 + 0.25 \text{ compact} + 0.15 \text{ edge}.$$



最终的校准置信度为：

$$\text{conf}_{\text{new}} = \text{clamp}(\text{conf}_{\text{orig}} \cdot f, 0.001, 0.999).$$

校准后置信度低于用户定义阈值（例如 0.25）的框被丢弃。

**置信度校准的优势：** (i) 利用低级图像特征（梯度）来验证框内是否实际包含裂纹状边缘模式，抑制来自平淡背景或光滑斑块的假阳性。(ii) 形状项整合了期望的裂纹伸长性，进一步提升真阳性并减少虚假检测。(iii) 整个校准是纯启发式的，无需额外训练，且增加运行时可忽略。

### 3.3.2 拓扑引导的框融合（TGF）

即使在 NMS 和置信度校准之后，一条长的连续裂纹仍可能由几个不连通的框表示，因为检测器在低对比度区域的置信度会下降。这些框通常具有较低的 IoU（常小于 0.2），标准 NMS 无法合并它们。为解决这个问题，我们提出一种基于图的融合算法，使用专门为裂纹设计的宽松连通性判据。

令  $\mathcal{B} = \{B_i\}_{i=1}^M$  为通过置信度校准的框集合。我们在  $\mathcal{B}$  上构建并查集（不相交集合并）数据结构。两个框  $B_i$  和  $B_j$  若满足以下任一条件，则视为**连通**：

- **准则 A（裂纹专用低 IoU）：** 两者都属于裂纹类且  $\text{IoU}(B_i, B_j) > 0.12$ 。阈值 0.12 远低于标准 NMS 阈值，允许合并轻微重叠的片段。
- **准则 B（基于方向的邻近性）：** 两者均为裂纹类，方向相同（若  $w \geq h$  则为水平，否则垂直），且中心欧氏距离  $d_{ij}$  满足：

$$d_{ij} < \max(0.08 \cdot D_{\text{img}}, 0.65 \cdot \max(L_i, L_j)),$$

其中  $D_{\text{img}}$  是图像对角线长度， $L_i$  是框  $B_i$  的较长边。该准则捕获沿同一方向对齐但不重叠的片段。

- **准则 C（非裂纹默认）：** 两者都属于非裂纹类且  $\text{IoU} > 0.35$ （标准阈值）。

构建连通图后，我们识别连通分量。对于每个分量  $C = \{B_{i_1}, B_{i_2}, \dots, B_{i_m}\}$ ，将框融合为单个检测，融合方式如下：

- **融合坐标：** 计算框坐标的加权平均（权重为校准后的置信度），同时也计算轴对齐边界包络（所有角点的最小/最大值）。最终坐标是两者的凸组合：

$$\begin{aligned} \text{xy}_w &= \frac{\sum_{k=1}^m \text{conf}_k \cdot (x_{1,k}, y_{1,k}, x_{2,k}, y_{2,k})}{\sum_{k=1}^m \text{conf}_k}, \\ \text{xy}_e &= \left( \min_k x_{1,k}, \min_k y_{1,k}, \max_k x_{2,k}, \max_k y_{2,k} \right), \\ \text{xy}_{\text{fused}} &= 0.35 \cdot \text{xy}_w + 0.65 \cdot \text{xy}_e. \end{aligned}$$

包络项（65% 权重）确保融合框覆盖裂纹的整个跨度，而加权项（35%）在高置信度片段时提供更精确的中心。

- **融合置信度：**取分量内置信度的均值，并增加一个多片段的小奖励：

$$\text{conf}_{\text{fused}} = \text{clamp}\left(\frac{1}{m} \sum_{k=1}^m \text{conf}_k + 0.05 \cdot \ln(1 + m), 0.001, 0.999\right).$$

对数奖励鼓励合并，因为一个被多个片段检测到的裂纹很可能是真阳性。

- **融合类别：**取分量中任意框的类别标签（根据构造它们应全相同；若出现冲突，取总置信度最高的类别）。

融合所有分量后，输出是融合后的框以及任何未合并的单框。此融合集替换原始检测结果。

**拓扑引导框融合的优势：** (i) 显式建模了单个物理裂纹可能在检测输出中被分割成多个碎片这一事实，这是基于 CNN 的检测器对细长物体常见的失败模式。(ii) 宽松的连通性判据（低 IoU 和基于方向的距离）专为裂纹定制，而非裂纹物体则回退到标准 NMS。(iii) 融合策略保留了裂纹的整体范围，产生单一的、连贯的检测，这对下游应用（如裂纹长度估计）更有用。

### 3.4 训练与推理设置

所有模型使用 SGD 优化器，动量为 0.937，权重衰减为 0.0005，初始学习率为 0.003，采用余弦退火调度，批量大小为 64（或在有限 GPU 上为 8），训练 300 轮。数据增强包括 mosaic（概率 0.6）、mixup（0.05）、随机缩放（±40%）和 HSV 偏移。应用早停，耐心为 80。输入图像尺寸为 640×640。所有实验在单个 NVIDIA Quadro P2000（或类似显卡）和 PyTorch 1.13 上进行。